

A 200g ball rotates at a constant speed of 3 m/s on the end of 1.2 m long string. The

string describes a horizontal circle.

What is the centripetal acceleration of the object in m/s^2 ?

What is the centripetal force exerted on the object in N?

Select one:

- ☐ a.
 $3.33 \text{ m/s}^2, 0.67 \text{ N}$
- ☒ b.
 $13.33 \text{ m/s}^2, 2.67 \text{ N}$
- ☐ c.
 $7.5 \text{ m/s}^2, 1.5 \text{ N}$
- ☐ d.
 $20.83 \text{ m/s}^2, 4.17 \text{ N}$

A simple pendulum of length L and mass M has frequency f . To increase its frequency to $5f$:

Select one:

☐ a.

Increase its length to $5L$

☐ b.

Increase its length to $25L$

☒ c.

Decrease its length to $L/5$

☐ d.

Decrease its length to $L/25$

A Ferrari is traveling in a uniform circular motion around a racetrack. What happens to the radial acceleration of the car if the velocity is doubled and the radius of the circle is doubled?

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Select one:

☐ a.

It increases by a factor of 4.

☒ b.

It increases by a factor of 2.

☐ c.

It remains the same.

☐ d.

It increases by a factor of 8.

Find the kinetic Energy of the ball
(Mass = 3000g) when it is 8m
above the ground in Joule:



Select one:



a.

240



b.

210



c.

225



d.

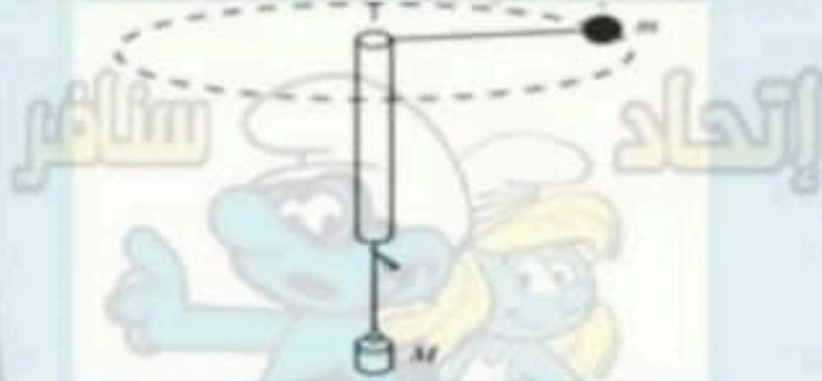
450

The percentage error of the rotational mass:

The actual value of the rotational mass equals 14g, and m complete 10 revolutions

in 5 second, $M = 75\text{g}$, $r = 30\text{ cm}$, $g = 10\text{ m/s}^2$.

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Select one:



a.

11.4%



b.

12.86%



c.

9.2%



d.

15.3%

A person exert a force on a string attached to a 325g ball to make the ball revolve

uniformly in a horizontal circle of radius 30cm. Find the tension in the string(in N) if

the ball makes two revolutions per second.

Select one:

☐ a.

13.8

☐ b.

15.38

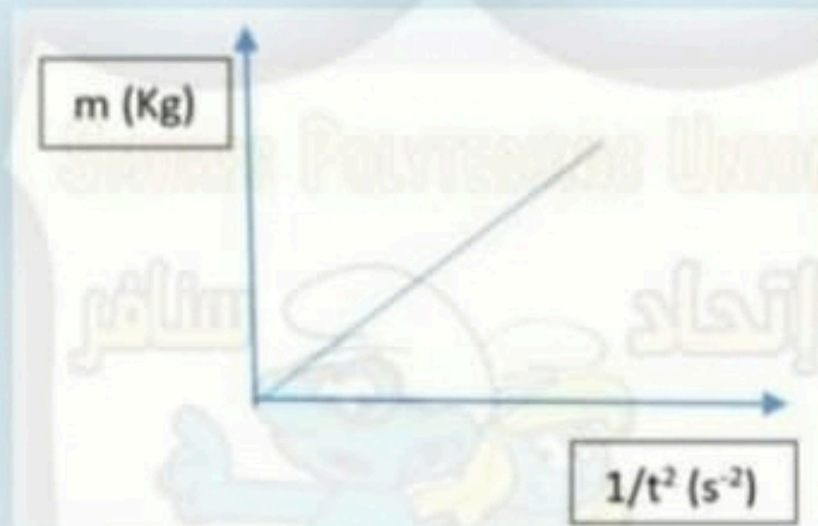
☒ c.

11.83

☐ d.

18.93

IN Newton's second law
Experiment, the total mass equals
250g, the distance between
light gates equals 70 cm, Find
acceleration of gravity in m/s^2 :
(Slope = 0.041 Kg.s^2)



Select one:

☒ a.

8.5

☐ b.

10.4

☐ c.

9.8

☐ d.

9.2

The period of vertical oscillation of a mass-spring system is T when the spring carries

a mass of 1.00 kg . What mass in (Kg) should be added to the 1.00 kg if the period is to

be increased to $1.50 T$?

Select one:

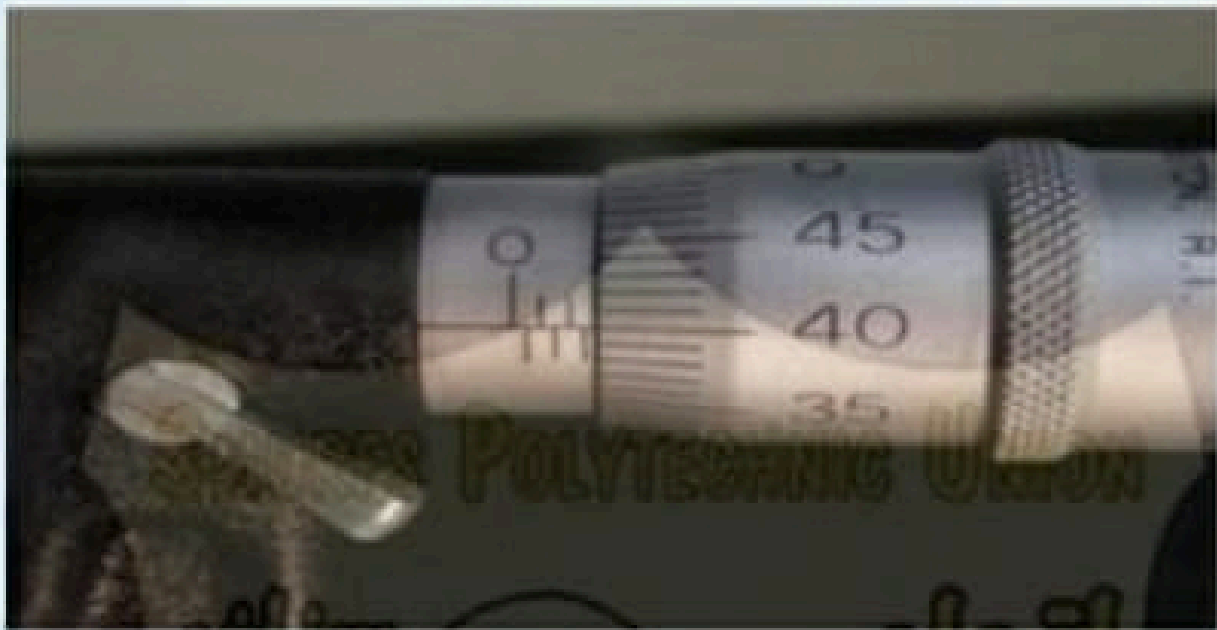
☐ a.
1.50

☐ b.
1.25

☐ c.
2.25

☒ d.
1.00

What does this metric micrometer read?



Select one:

☒ a.

2.90 mm

☐ b.

0.29 cm

☐ c.

2.40 mm

☐ d.

5.40 mm

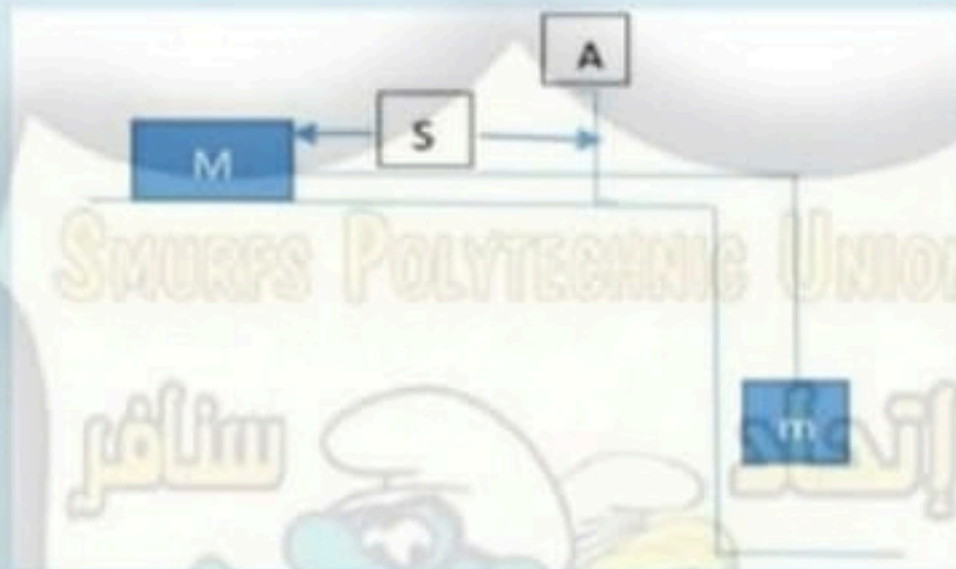
A simple pendulum is moved from the Earth ($g = 10 \text{ m/s}^2$) to the X planet. How does it

change the period of oscillations?
Acceleration due to gravity on X planet = $(5/3) \text{ m/s}^2$:

Select one:

- ☐ a. The period is increased by factor $\sqrt{6}$
- ☐ b. The period is decreased by factor $\sqrt{6}$
- ☐ c. The period is decreased by factor four
- ☐ d. The period is increased by factor four
- ☒ e. The period remains the same

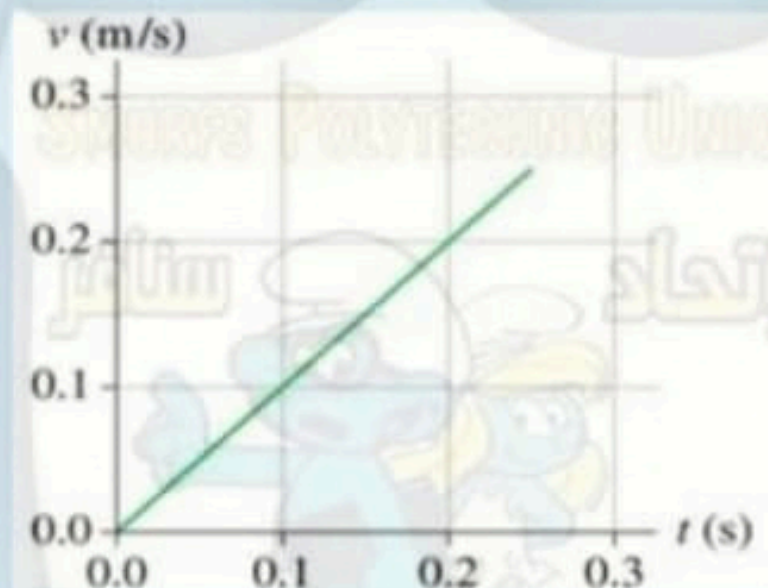
In G.P.E. Experiment, $M = 350\text{g}$, $m = 40\text{g}$, $L = 2\text{ cm}$, $t_A = 24.6\text{ ms}$, Find S in cm:



Select one:

- ☐ a.
21.8
- ☐ b.
28.1
- ☐ c.
32.2
- ☒ d.
64.4

Scallops eject water from their shells to provide a thrust force. The graph shows a smoothed graph of actual data for the initial motion of a 25 g scallop speeding up to escape a predator. What is the magnitude of the net force(N) needed to achieve this motion?



Select one:

☒ a.

0.025

☐ b.

05

☐ c.

2.5

☐ d.

0.005

A 23 kg cart (A) moving at 8 m/s collides with a 16 kg cart (B) moving 13 m/s. If the

carts stick together after the collision, what is the velocity after the collision?



Select one:

- ☐ a.
- 1.04 m/s
- ☒ b.
- 0.62 m/s
- ☐ c.
0.6 m/s
- ☐ d.
10.05 m/s

A 1250 kg car is moving down the highway with a velocity of 38.0 m/s when it bumps into the car ahead of

its which has a mass of 900 kg and a velocity of 30.0 m/s. After the collision, the two cars stick together.

What will be the resulting velocity of the two cars together? How much energy will be lost in this collision?

Select one:

☒ a.

(32.7 m/s, 16879.4 J)

☐ b.

(29.1 m/s, 13702 J)

☐ c.

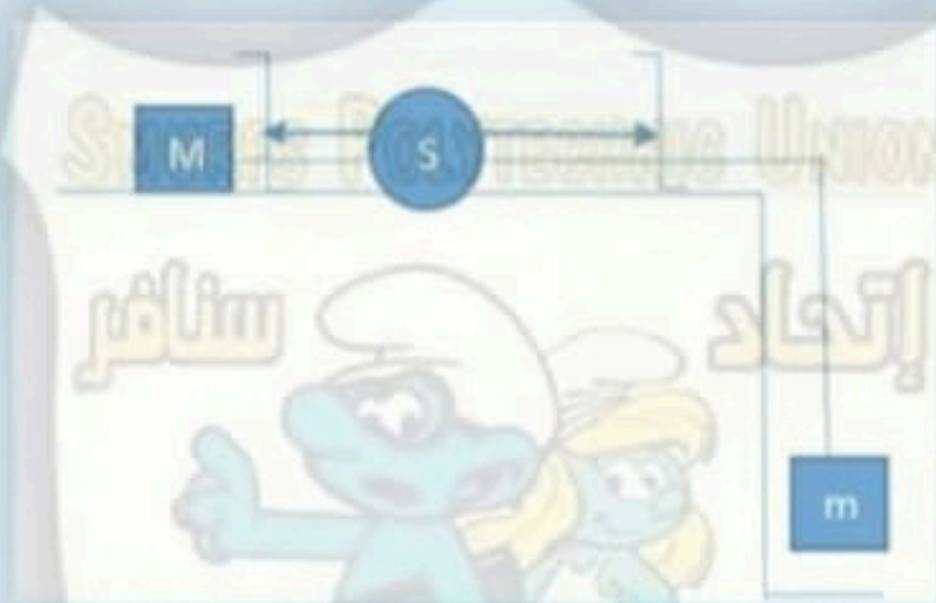
(34.65 m/s, 16830.8 J)

☐ d.

(30.2 m/s, 23307 J)

The time required to cut the distance between the two photogates in second:

($M = 220 \text{ g}$, $m = 30 \text{ g}$, $S = 70 \text{ cm}$, $g = 10 \text{ m/s}^2$)



Select one:

- ☐ a.
1.97
- ☐ b.
1.08
- ☒ c.
1.33
- ☐ d.
0.95

A 2500 N Car is supported by 5 springs. If each spring is deflected by 4 cm what is the spring constant in (N/m)?

Select one:

☒ a. 62500

☐ b. 500

☐ c. 12500

☐ d. 125

Three simple pendulums, with masses m_1 , $m_2 = 2m_1$, and $m_3 = 3m_1$ hung at the end,

have the same shape and size and are suspended at the same point. Rank them

according to their periods, from shortest to longest.

Select one:



a.

All three are the same



b.

2, 1, 3



c.

3, 2, 1



d.

1, 2, 3

An object of mass $m_1 = 4 \text{ Kg}$, moving with velocity $V_{i1} = 12 \text{ m/s}$, collides head-on with a stationary object

whose mass is $m_2 = 8 \text{ Kg}$. Given that the collision is elastic, what are the final velocities of the two objects

(V_{1f}, V_{2f})? Neglect friction.

Select one:

☐ a.

$(-6, 6) \text{ m/s}$

☒ b.

$(-4, 8) \text{ m/s}$

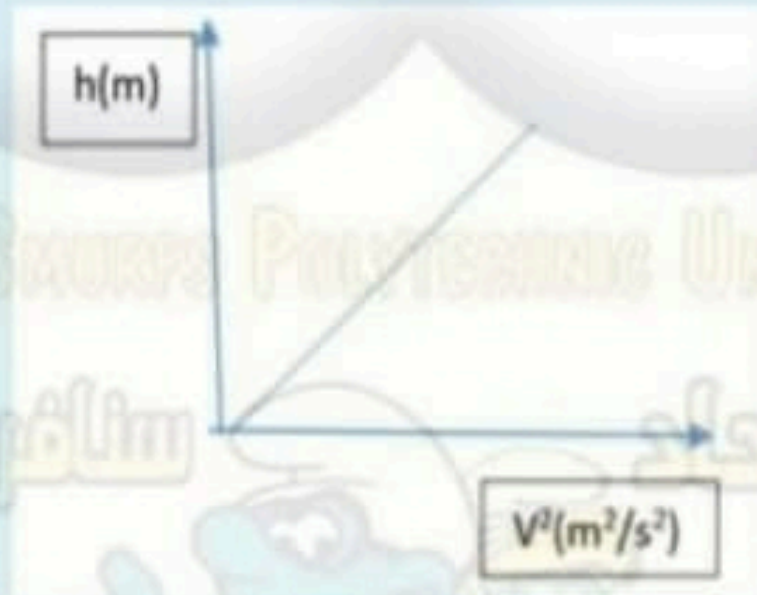
☐ c.

$(-12, 12) \text{ m/s}$

☐ d.

$(7.2, 2.4) \text{ m/s}$

In G.P.E. Experiment 2, $M = 200\text{g}$, $m = 20\text{g}$, Slope = $0.65 \text{ s}^2/\text{m}$, Find the acceleration of gravity in (m/s^2) :



Select one:



a.

12



b.

9.8



c.

11



d.

8.87

Find:

$$4.7832 + 1.234 + 2.02 =$$

(Using significant figure rules)

Select one:



a.

8.04



b.

8.0372



c.

8.037



d.

8.0

A cylinder hangs motionless 20 cm below its equilibrium position from a spring with a

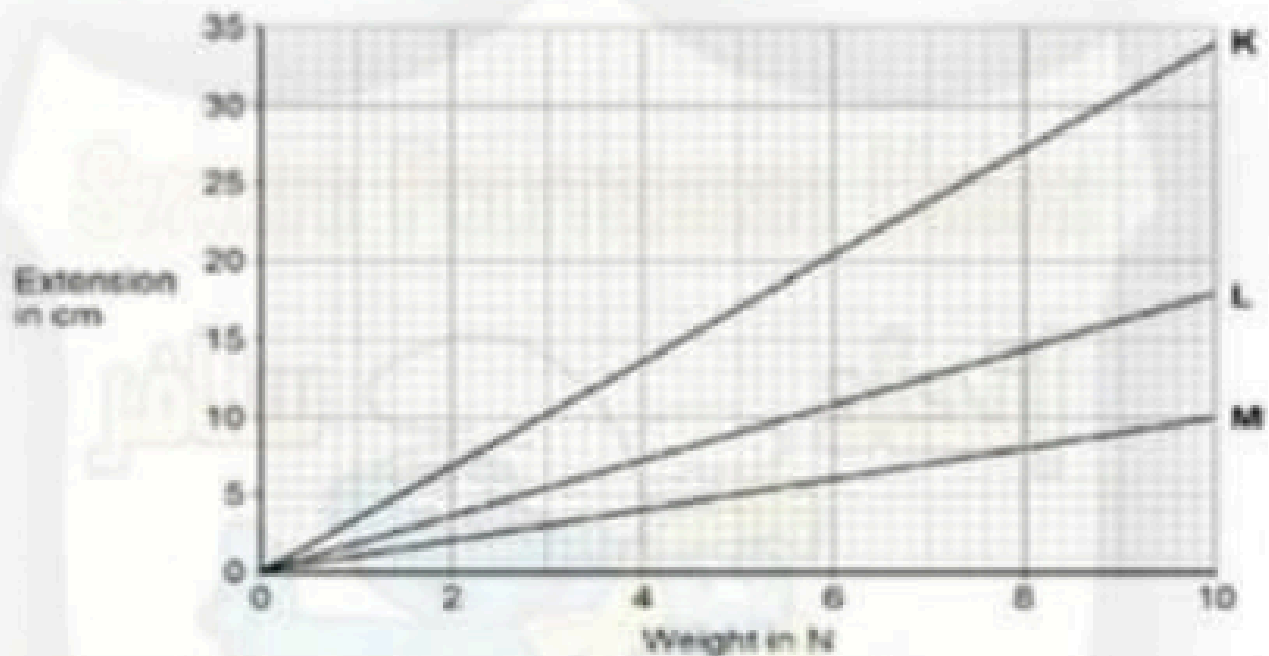
constant of 50 N/m. What is the mass of the cylinder?

Select one:

- ☐ a.
1
- ☐ b.
10
- ☒ c.
100
- ☐ d.
25



Which out of springs K, L or M from the graph have the highest spring constant.



Select one:



a.

M



b.

Same spring constant



c.

K

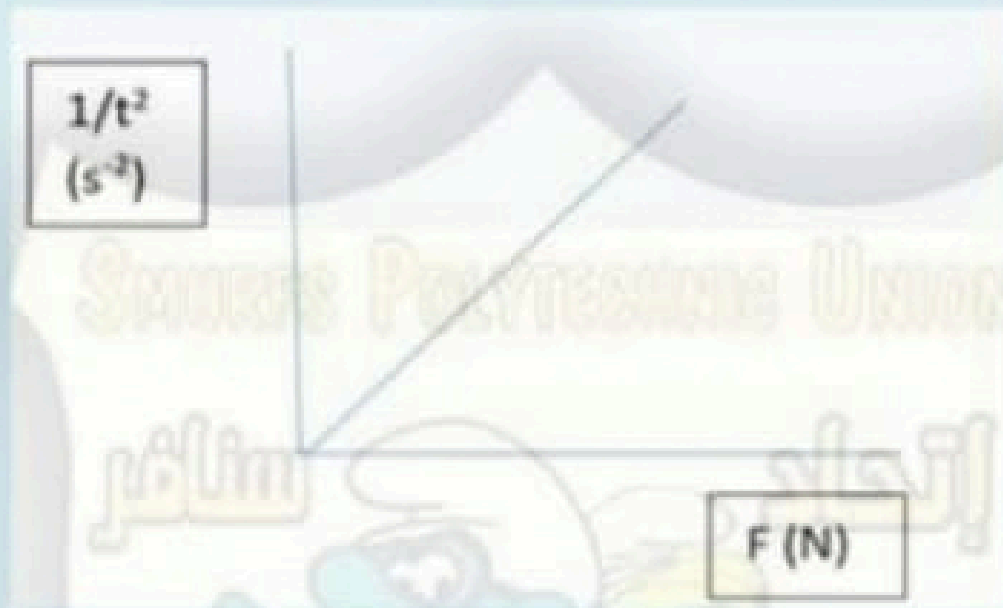


d.

L

Find the total mass of the system:

($S = 75.2 \text{ m}$, $g = 10 \text{ m/s}^2$, Slope = $3.03 \text{ s}^{-1}/\text{N}$)



Select one:



a.

150



b.

250



c.

220



d.

180